ME 264 Mechanisms Synthesis & Analysis I

Unit 5 Rotary motion Transmission

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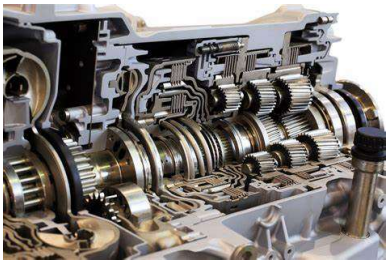
1

Introduction

- Rotary motions are general transmitted from one shaft to another using rollers, gears, belts, and chain drives.
- Rollers, belts and ropes use friction as useful agent to transmit power, but are subjected to uncertainties in the values of coefficient of friction.
- In general, gears and chains transmit power efficiently by eliminating slipping.

2

Examples of Drives



Gear Drive

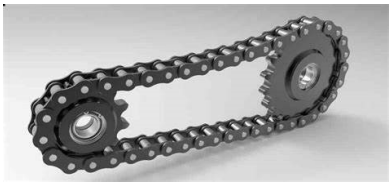




Timing Belt Drive



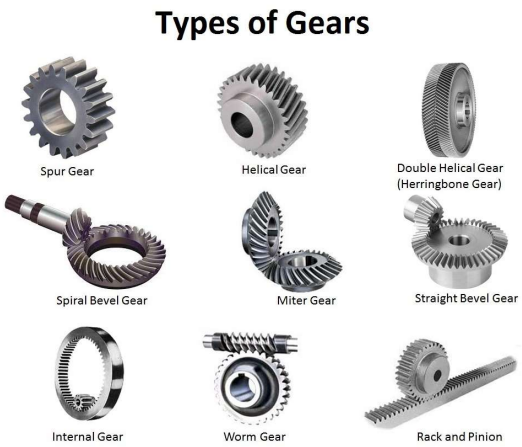
Belt Drive



Chain Drive

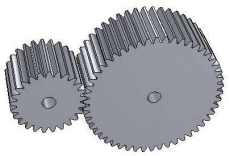
Types of Gears

- Gears may be divided into three main groups, which are:
  - parallel-axis gears
  - non-parallel-axis co-planer gears
  - Non-parallel-axis and non-coplaner gears.



Type of Gears: Parallel Axis Gears

- Parallel-axis gears are the simplest and most common type of all gear types.
- They include spur and helical gears.
- They can transmit large tongue with high efficiency.





Spur gear





Helical gear

5

Type of Gears: non-parallel-axis co-planer gears Gears

- These gears have nonparallel axis of rotation unlike spur and helical gears, but are coplanar.
- Bevel and spiral gears are examples.
- The common feature of this group of gears is redirection of power from one axis to another, though the two pair of gears (input and output) which are not parallel but coplanar (same xy-plane).



Bevel gear



Spiral toothed Bevel gear



Spiral toothed Bevel gear

6

Type of Gears: non-parallel-axis non-coplaner gears

- These gears are more complex in both geometry and manufacturing than parallel-axis gears.
- As such, these gears are more expensive than parallel-axis and nonparallel coplanar gears.
- They include worm gear drives



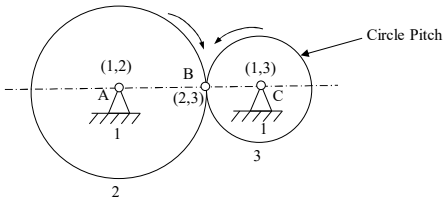


General Rotation Train Equation

- Shown are two roller discs 2 and 3 having a common contact point at B.
- The instant centres are (1, 2), (2, 3) and (1, 3).
- From the Kennedy's theorem, the three instant centres lie on the same line. This implies that the line, which passes through the centres of the discs, must be the common normal to both discs.
- For continuity, the velocity of the common point of contact is given by:

$$\omega_2 r_{B/A} = \omega_3 r_{C/B}$$

$$\frac{\omega_2}{\omega_3} = \frac{r_{C/B}}{r_{B/A}} = \frac{\left|IC_{(2,3)} - IC_{(1,3)}\right|}{\left|IC_{(2,3)} - IC_{(1,2)}\right|} \quad \text{----(1)}$$



General Rotation Train Equation

- Since the instant centres lie on the same line, Equation (1) is negative and therefore the two gears rotate in opposite directions.

- The equation also shows that the angular velocity ratio ω_2/ω_3 is inversely proportional to the pitch-radius ratio r_3/r_2, the pitch-diameter ratio d_3/d_2, the pitch-circumference ratio c_3/c_2, or the number of teeth N_3/N_2. Thus

$$\pm\frac{\omega_2}{\omega_3}=\frac{r_3}{r_2}=\frac{d_3}{d_2}=\frac{c_3}{c_2}=\frac{N_3}{N_2} \qquad \text{----------(2)}$$

where sign = - if the two engaging gears are external gears and, sign = + if one of the engaging gears is external (ring, internal or annular) gear.

9

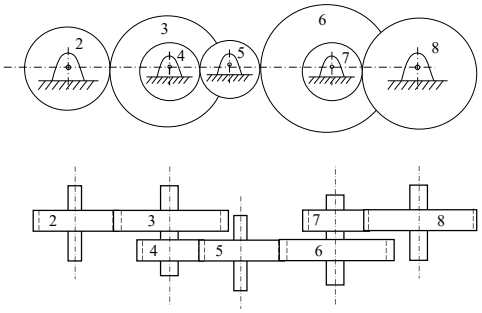
Ordinary Gear Train

- Sometimes, a design may call for a high angular velocity ratio that will be impractical to achieve due to space and material strength limitations.

- An alternate way of achieving such large angular velocity ratio is to use a series *of pair of gear* set or ordinary gear.

- Ordinary gear train consists of at least two pairs of meshing gears, where the output angular velocity of one pair of gears is coupled to the input of the next pair of gears. Thus, the velocity ratio is changed at several stages.

Find the velocity ratio ω_8/ω_2 of the gear trains shown.



10

Ordinary Gear Train

The first stage engagement is between gears 2 and 3. Gears 3 and 4 are keyed one shaft, and so are gears 6 and 7. From Equation (2) and knowing that both gears are external gears.

$$\omega_8 = \omega_7\left(-\frac{N_7}{N_8}\right) \quad \text{----------(1)}$$

Where Sgn = -1. Since gears 6 and 7 are rigidly attached to the same shaft, $\omega_6 = \omega_7$. Thus

$$\omega_8 = \omega_6\left(-\frac{N_7}{N_8}\right) \quad \text{----------(2)}$$

Similarly, $\omega_6 = \omega_5\left(-\frac{N_5}{N_6}\right)$ ----------(3)

Substituting equation (3) into (2) yields

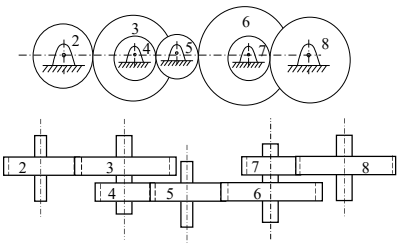
$$\omega_8 = \left[\omega_5\left(-\frac{N_5}{N_6}\right)\right]\left(-\frac{N_7}{N_8}\right) \quad \text{----------(4)}$$

Similarly $\omega_5 = \left[\omega_2\left(-\frac{N_2}{N_3}\right)\right]\left(-\frac{N_4}{N_5}\right)$ ----------(5)

Replacing ω_5 in Equation (5) with equation (5), and dividing through by ω_2 yields

$$\frac{\omega_8}{\omega_2} = \left(-\frac{N_2}{N_3}\right)\left(-\frac{N_4}{N_5}\right)\left(-\frac{N_5}{N_6}\right)\left(-\frac{N_7}{N_8}\right) \quad \text{-------(6)}$$

$$\frac{\omega_8}{\omega_2} = \left(\frac{N_2}{N_3}\right)\left(\frac{N_4}{N_5}\right)\left(\frac{N_5}{N_6}\right)\left(\frac{N_7}{N_8}\right)(-1)^4 \quad \text{------(7)}$$



General Ordinary Gear Train Equation

- Equations (6) and (7) may be generalised as

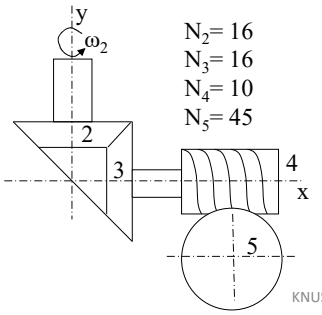
$$\frac{\omega_{driven}}{\omega_{driver}} = \frac{\text{product of number of teeth on driver gears}}{\text{product of number of teeth on driven gears}} \quad \text{--(8)}$$

$$\frac{\omega_{driven}}{\omega_{driver}} = \frac{\text{product of number of teeth on driver gears}}{\text{product of number of teeth on driven gears}} x(-1)^n \quad \text{--(9)}$$

- where n is number of external meshes (or engagements).
- In Equation (8), the negative sign that accounts for direction of rotation is omitted, but it must be accounted for when using it.
- If an external gear meshes with an internal (annular or ring), the negative sign is omitted. Equation (8) is applicable to all types of gears, belt drives, chain drives and rollers but Equation (9) is applicable to only gears.
- In belt and chain drives, the number of teeth (N) is replaced with the diameter of pulley and number of sprockets, respectively.
- In equation (6), N_5's cancels out but the negative sign remains. This implies that the presence of gear 5 does not affect the magnitude of the angular velocity ratio, but it causes direction reversal. Gear 5 is known as ***idler***, and its main purpose is to change the direction of rotation.

Example 5:2

The gear train shown consists of miter (same-size bevel) gears having 16 teeth each, a 10-tooth right hand worm and 45-tooth worm gear. If the input bevel gear 2 rotates at ω_2 =150 rpm in the anti-clockwise direction, what is the magnitude of the speed of the worm gear 5.



$$\frac{\omega_5}{\omega_2} = \left(\frac{N_2}{N_3}\right)\left(\frac{N_4}{N_5}\right)$$

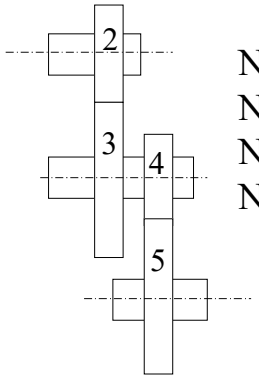
$$\frac{\omega_5}{\omega_2} = \left(\frac{16}{16}\right)\left(\frac{10}{45}\right) = {}^{2}/_{9}$$

$$\omega_5 = \frac{2}{9}\omega_2 = \frac{2}{9}(150)$$

$$\omega_5 = 33.3 \text{ rpm}$$

Example 5:1

Determine the angular velocity ratio ω_5/ω_2 of the gear train shown



$N_2 = 20$
$N_3 = 80$
$N_4 = 15$
$N_5 = 60$

$$\frac{\omega_5}{\omega_2} = \left(-\frac{N_2}{N_3}\right)\left(-\frac{N_4}{N_5}\right)$$

$$\frac{\omega_5}{\omega_2} = \left(\frac{20}{80}\right)\left(\frac{15}{60}\right) = {}^{1}/_{16}$$

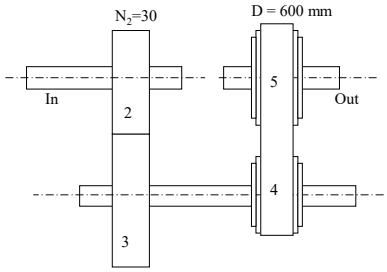
$$\frac{\omega_5}{\omega_2} = {}^{1}/_{16}$$

Example 5:3

Shown is a compound gear train, which the inlet gear 2 rotates at 1200 rpm in the anti-clockwise direction. If the angular velocity of outlet shaft is 300 rpm and the gear contact between gear 2 and gear 3 reduces the speed ratio by 5:3, find

(a) the number of teeth of gear 3,

(b) diameter of pulley 4.



(a) $\frac{\omega_3}{\omega_2} = \left(-\frac{N_2}{N_3}\right) = -\frac{3}{5}$

$N_3 = \frac{5}{3}N_2 = \frac{5}{3}(30) \quad N_3 = 50$

(b) $\frac{\omega_5}{\omega_2} = \left(-\frac{N_2}{N_3}\right)\left(\frac{D_4}{D_5}\right)$

$\omega_5 = \left(-\frac{N_2}{N_3}\right)\left(\frac{D_4}{D_5}\right)\omega_2$

$-300 = \left(-\frac{3}{5}\right)\left(\frac{D_4}{600}\right)1200$

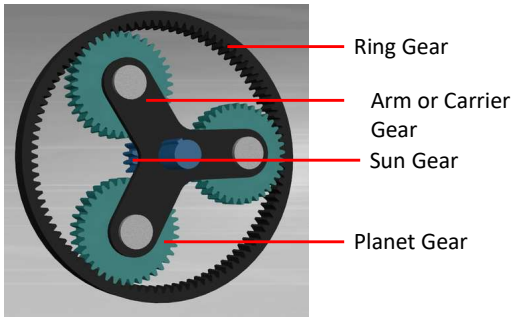
$D_3 = 250\ mm$

Note that we could solve the problem in absolute values by dropping the negative signs

15

Planetary (or Epicyclic) Gear Train

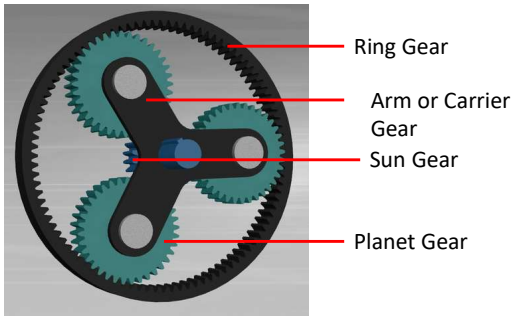
- Planetary gear trains are often utilised to make compact gear reducer.
- One major advantage of planetary gear train is that it has two degrees of freedom mechanism.
- Planetary gear train has domestic and industrial applications: domestic and industrial food mixers, automatic transmission, packaging of food products into carton boxes, and automated packaging papers and toilet rolls.



16

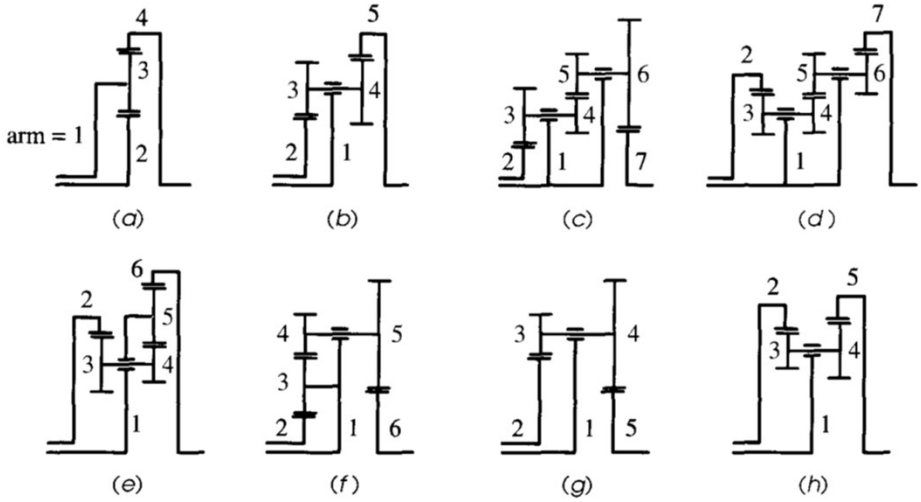
Planetary (or Epicyclic) Gear Train

- A typical planetary gear train consists a ring (or annular or external) gear, sun gear, planet gears and arm.
- The arm carries the planet gears and rotates with them. As the arm rotates, the planet gears engage both the ring gear and the sun gear.



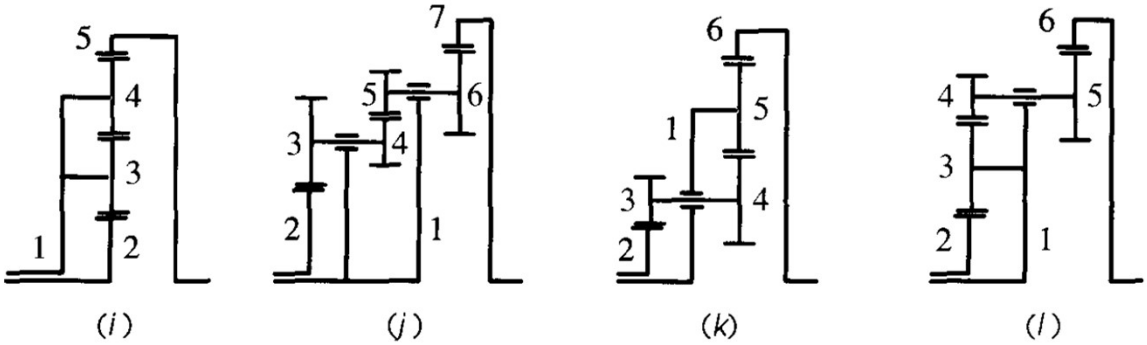
17

Levai's 12 Possible Planetary Trains

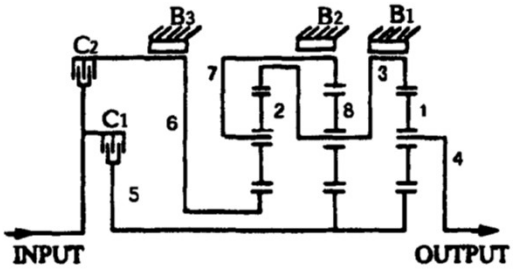


18

Levai's 12 Possible Planetary Trains



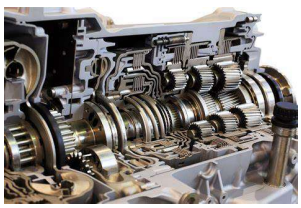
Schematic of 4-Speed Automatic Transmission



(a) Schematic of 4-speed automatic transmission

Range	Clutch/Brake Activation				
	C_1	C_2	B_1	B_2	B_3
First	x		x		
Second	x			x	
Third	x				x
Fourth	x	x			
Reverse		x	x		

(b) Clutch / brake activation table



Planetary Gear: One Input (Formular Method)

Consider a planetary train, which has no ring gear; sun gear 2 fixed to the ground and the arm 4 carries planet gear 3 while it rotates with angular velocity with respect to the ground

$$\omega_{31} = \omega_{41} + \omega_{34}$$

Dividing equation ω_{41}

$$\frac{\omega_{31}}{\omega_{41}} = 1 + \frac{\omega_{34}}{\omega_{41}}$$

Since the gear 2 is fixed to the ground

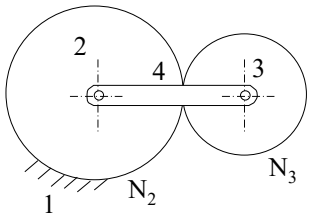
$$\omega_{41} = \omega_{42}$$

Replacing ω_{41} by ω_{42}

$$\frac{\omega_{31}}{\omega_{41}} = 1 + \frac{\omega_{34}}{\omega_{42}}$$

or

$$\frac{\omega_{31}}{\omega_{41}} = 1 - \frac{\omega_{34}}{\omega_{24}}$$



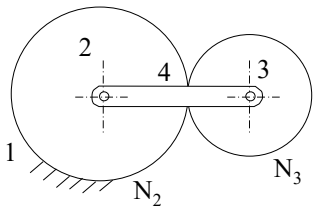
Planetary Gear: One Input (Formular Method)

- The angular velocity ratio ω_{34}/ω_{24} is the angular velocity ratio with respect to the arm.
- The arm may be considered as fixed and, hence, treat the gear train as an ordinary gear train in this kinematic inversion

$$\frac{\omega_{31}}{\omega_{41}} = 1 - \left(-\frac{N_2}{N_3}\right)$$

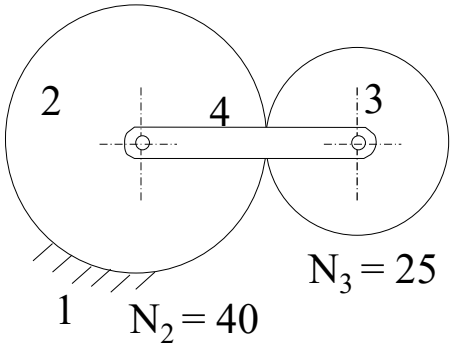
$$\frac{\omega_{31}}{\omega_{41}} = 1 + \frac{N_2}{N_3}$$

$$\frac{\omega_{31}}{\omega_{41}} = 1 + \frac{d_2}{d_3}$$



Example 5:4

If the arm 4 has an anti-clockwise velocity of 4 rad/s, determine the absolute angular velocity of gear 3.



$$\omega_{31} = \left(1 + \frac{N_2}{N_3}\right)\omega_{41}$$

$$\omega_{31} = \left(1 + \frac{40}{25}\right)4$$

$$\omega_{31} = 10.4\, rad/s$$

Planetary Gear: One Input (Formular Method)

Consider a planetary train, which has no ring gear and has sun gear 2 and arm 4 are the two inputs. Then, the absolute velocity of planet gear 3 and sun gear 2 are,

$$\omega_{34} = \omega_{31} - \omega_{41} \quad (1)$$

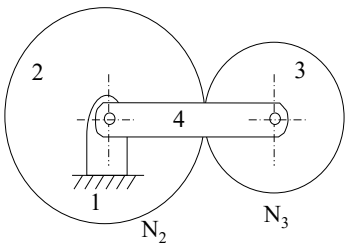
$$\omega_{24} = \omega_{21} - \omega_{41} \quad (2)$$

Dividing equation (1) by equation (2) gives

$$\frac{\omega_{34}}{\omega_{24}} = \frac{\omega_{31} - \omega_{41}}{\omega_{21} - \omega_{41}} \quad (3)$$

If the arm 4 is fixed, then

$$\frac{\omega_{31} - \omega_{41}}{\omega_{21} - \omega_{41}} = \frac{\omega_{34}}{\omega_{24}} = -\frac{N_2}{N_3} \quad (4)$$



Planetary Gear: One Input (Formular Method)

Equation (4) may be generalized as

$$\frac{\omega_{LA}}{\omega_{FA}} = \frac{\omega_L - \omega_A}{\omega_F - \omega_A} = \frac{P}{Q} \qquad (5)$$

ω_{LA} = velocity of the last gear relative to that of the arm

ω_{FA} = angular velocity of the first gear relative to that of the arm

ω_L =absolute angular velocity of the last gear

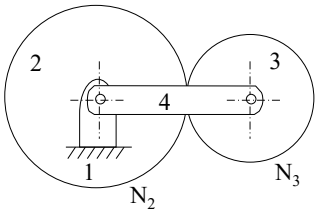
ω_F =absolute angular velocity of the first gear

ω_A =absolute angular velocity of the first arm

P = product of number of teeth on driver gears

Q = product of number of teeth on driven gears

Note: Sign should be accounted for external engagements. Equation (5) is applicable to all gear types, chain drives, rollers and belts drives.



25

Example 5:5

Find N4 in terms of N2 and N3 if the gears have same module.

Solution

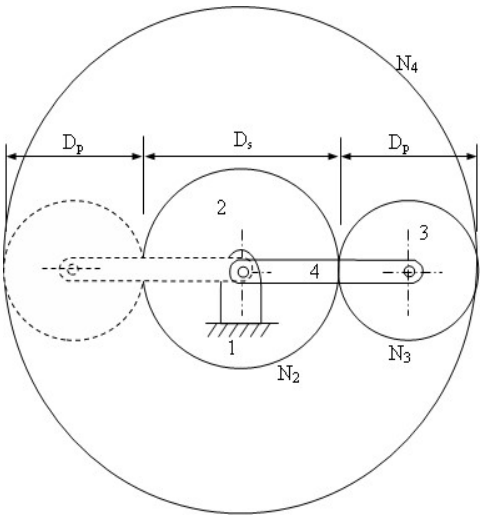
$D_p+D_s+D_p=D_r$

$D \alpha N$

Hence

$2N_p+N_s=N_s$

$N_4=2N_2+N_3$



26

Examples 5:6

Find the number of teeth N_4 on gear

4 if N_2=40 and N_3=30

Solution

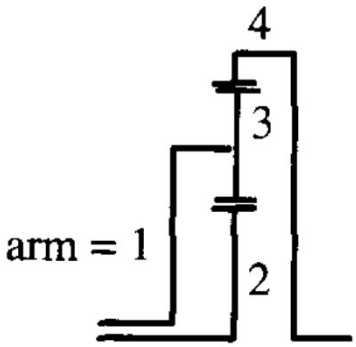
$$R_4=R_2+2R_3$$

since $R \alpha N$

$N_4 = N_2+2N_3$ 40+2(30)

$N_4 = 40+2(30)$

$N_4 = 100$



Examples 5:7

Find the number of teeth N_5 on gear 5

of if N_2=40, N_3=20 and N_4= 30

Solution

$R_5=R_2+R_3+R_4$

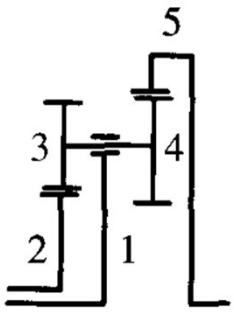
Since $R \alpha N$

Hence

$N_5 = N_2+N_3+N_4$

$N_5 = 40+20+30$

$N_5 = 90$



Examples 5:8

Find the number of teeth N_6 on gear 6 if

N_2=30, N_3=40, N_4=25 and N_5=85

Solution

$R_2+2R_3+R_4=R_5+R_6$

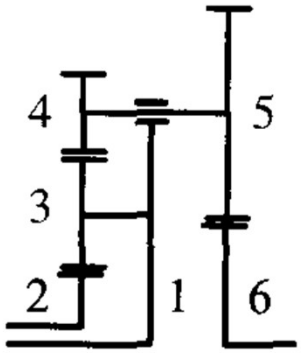
$R \alpha N$

Hence

$N_2+2N_3+N_4=N_5+N_6$

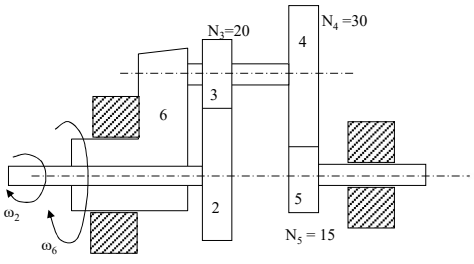
$30+2(40)+25=85+N_6$

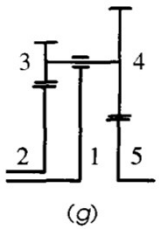
$N_6=50$



Example 5:9

Shown is planetary gear train with sun gear 2 and arm 6 as inputs. Determine the angular velocity of sun gear 5 if sun gear 2 rotates at 50 rad/s and arm 6 rotates at 75 rad/s, both in anti-clockwise as viewed from the left.





(g)

Solution

Considering sun gear 2 as first (driver) and sun gear 5 as last (driven) gear

$N_2 + N_3 = N_4 + N_5$ $\qquad N_2 = N_4 + N_5 - N_3$

$N_2 = 30 + 15 - 20$ $\qquad N_5 = 25$

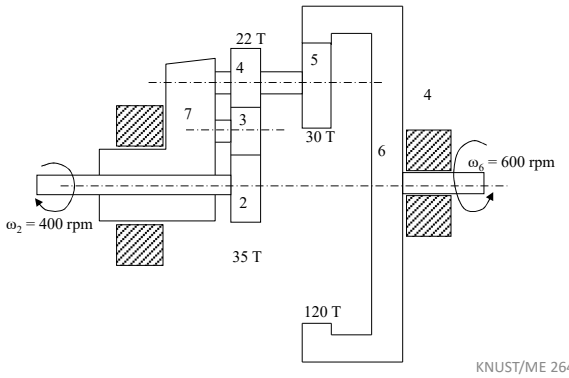
$$\frac{\omega_{56}}{\omega_{26}} = \frac{\omega_5 - \omega_6}{\omega_2 - \omega_6} = \left(-\frac{N_2}{N_3}\right)\left(-\frac{N_4}{N_5}\right)$$

$$\frac{\omega_5 - 75}{50 - 75} = \left(-\frac{25}{20}\right)\left(-\frac{30}{15}\right)$$

$\omega_5 = 12.5$ rad/s

Example 5:10

The gear train shown has two inputs: sun gear 2 rotates at 400 rpm and ring gear 6 rotates at 600 rpm, both in anti-clockwise as viewed from the right. Find the resulting angular velocity of arm 7



Solution

Considering sun gear 2 as first (driver) and sun gear 6 as last (driven) gear

$$\frac{\omega_{67}}{\omega_{27}} = \frac{\omega_6 - \omega_7}{\omega_2 - \omega_7} = \left(-\frac{N_2}{N_3}\right)\left(-\frac{N_3}{N_4}\right)\left(\frac{N_5}{N_6}\right) = \left(\frac{N_2 N_5}{N_4 N_6}\right)$$

$$\frac{600 - \omega_7}{400 - \omega_7} = \frac{(35)(30)}{(22)(120)}$$

$$\omega_5 = 732\ rpm$$

Tabular Method

Tabular method is based on kinematic inversion. It involves analysing two easily described parts of the motion separately, and then adding the results together. The following steps should be used:

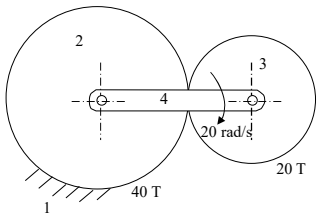
(1) Disconnect any gears from ground (if any gear/arm is fixed to the ground), and fixed all gears rigidly to the rotating arm

(2) Motion with arm: Rotate the arm with rigidly attached gears by a number of revolutions proportional to the angular velocity of the arm.

(3) Motion relative to arm: Separate the gears from the arm while holding the arm fixed, and rotate the rest of the gears back so that the total rotation (step 2 + step 3) of one or more of the gears matches their given rotations.

Example 5:11

Find the angular velocity of planet gear 3 if arm 4 rotates at 20 rad/s in clockwise direction.

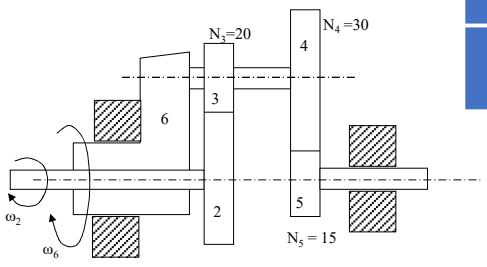


	Gear 2	Gear 3	Arm 4
Motion with arm	20	20	20
Motion relative to arm	-20	+20(N_2/N_3)	0
Total Motion	0	20+20(N_2/N_3)	20

$$\omega_3 = 20 + 20\left(\frac{N_2}{N_3}\right) = 20 + 20\left(\frac{40}{20}\right) = 60 \quad \text{rad/s}$$

Example 5:12 (same to Examples 5:9 & 5:12)

Shown is planetary gear train with sun gear 2 and arm 6 as inputs. Determine the angular velocity of sun gear 5 if sun gear 2 rotates at 50 rad/s in and arm 6 rotates at 75 rad/s, both in anti-clockwise as viewed from the left.



Work Direction

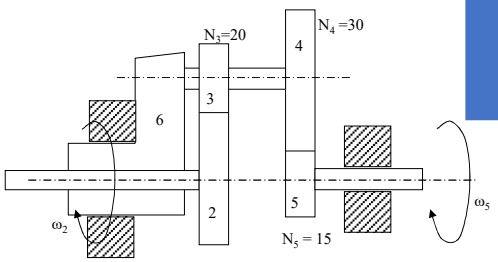
	Gear 2	Gear 3	Gear 4	Gear 5	Arm 4
Motion with arm	75	75	75	75	75
Motion relative to arm	-25	-25(-N_2/N_3)	25(N_2/N_3)	25(N_2/N_3) (-N_4/N_5)	0
Total Motion	50			75-25(N_2/N_3) (N_4/N_5)	75

$$\omega_5 = 75 - 25\left(\frac{N_2}{N_3}\right)\left(\frac{N_4}{N_5}\right) \qquad \omega_5 = 75 - 25\left(\frac{20}{16}\right)\left(\frac{30}{15}\right)$$

ω_5 =12.5 rad/s

Example 5:13

Shown is planetary gear train with sun gear 2 and sun 5 as inputs. Determine the angular velocity of arm 6 if sun gear 2 rotates at 50 rad/s and sun gear 5 rotates at 75 rad/s, both in anti-clockwise as viewed from the left.



Work Direction →

	Gear 2	Gear 3	Gear 4	Gear 5	Arm 4
Motion with arm	x	x	x	x	x
Motion relative to arm	50-x	(50-x)(-N_2/N_3)	(50-x)(-N_2/N_3)	(50-x)(-N_2/N_3)(-N_4/N_5)	0
Total Motion	50			x+(50-x)(-N_2/N_3)(-N_4/N_5)	x

$x+(50-x)(-N_2/N_3)(-N_4/N_5)=75$

$x+(50-x)(-20/16)(-30/15)=75$

$X=33.33$ rpm

35

Example 5.14

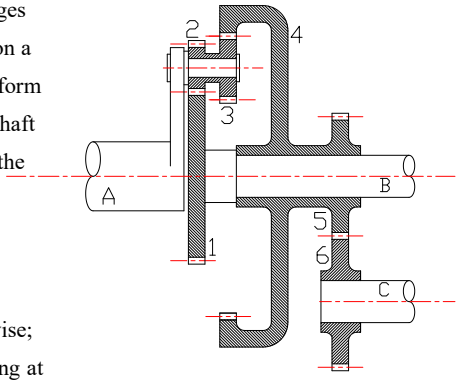
Two shafts A and B are co-axial, and are parallel to shaft C. Gear 1 is keyed to shaft B. Gears 2 and 3 form a compound gear and engages gears 1 and internal gear 4. The compound gear 2-3 is mounted on a pin which is carried by a carrier keyed to shaft A. Gears 4 and 5 form a compound gear, and gear 5 engages gear 6, which is keyed to shaft C. All the gears have the same module. The number of teeth on the gears are: $N_1 = 50$, $N_2=20$, $N_3 = 35$, $N_5=30$ and $N_6 = 40$.

(a) Find the number of teeth on gear 4.

(b) Determine the speed of shaft C, if

(i) shaft A is fixed and shaft B is rotating at 110 rpm anticlockwise;

(ii) shaft A is rotating at 75 rpm clockwise and shaft B is rotating at 110 rpm anticlockwise.



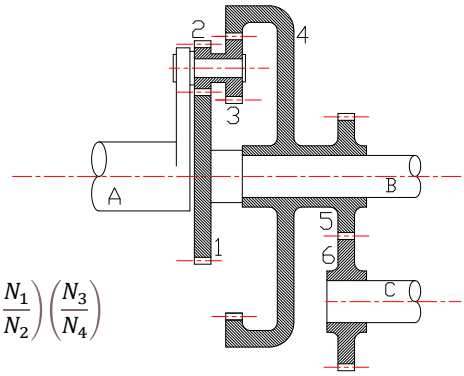
36

Example 5.14 conti;

(a) $R_1 + R_2 + R_3 = R_4$

For same module N α R

$N_4 = N_1 + N_2 + N_3 \qquad N_4 = 50 + 20 + 35 \qquad N_4 = 105$



(b) $\dfrac{\omega_4 - \omega_A}{\omega_1 - \omega_A} = \left(-\dfrac{N_1}{N_2}\right)\left(\dfrac{N_3}{N_4}\right) \quad \Rightarrow \omega_4 = \omega_A + (\omega_1 - \omega_A)\left(-\dfrac{N_1}{N_2}\right)\left(\dfrac{N_3}{N_4}\right)$

$\omega_4 = \omega_5 \qquad \omega_6 = \omega_5\left(-\dfrac{N_5}{N_6}\right)$

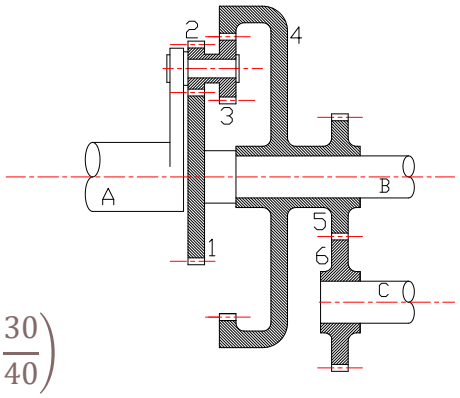
$\therefore \omega_C = \omega_6 = \left[\omega_A + (\omega_1 - \omega_A)\left(-\dfrac{N_1}{N_2}\right)\left(\dfrac{N_3}{N_4}\right)\right]\left(-\dfrac{N_5}{N_6}\right)$

Example 5.14 cont;

(b) (i) $\omega_A = 0 \; and \; \omega_1 = 110$ rpm

$\omega_C = \left[0 + (110 - 0)\left(-\dfrac{50}{20}\right)\left(\dfrac{35}{105}\right)\right]\left(-\dfrac{30}{40}\right)$

$\omega_C = 68.75$ rpm



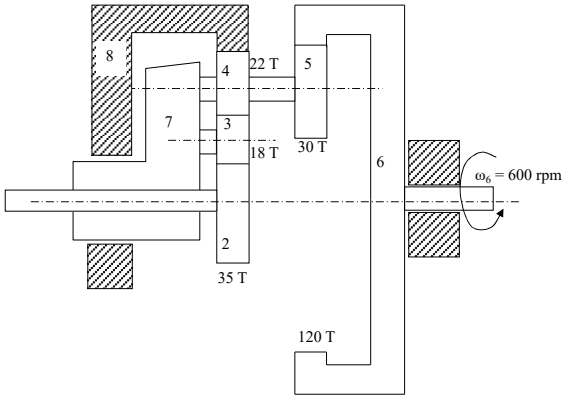
(ii) $\omega_A = -75 \; and \; \omega_1 = 110$ rpm

$\omega_C = \left[-75 + (110 - -75)\left(-\dfrac{50}{20}\right)\left(\dfrac{35}{105}\right)\right]\left(-\dfrac{30}{40}\right)$

$\omega_C = 132.5$ rpm

Example 5.15

Find the angular velocity of sun gear 2 if ring gear 6 rotates at 600 rpm in anti-clockwise as viewed from the right.



39

Example 5.15 Solution

- Here, two tables are created for paths from gear 6 to gear 8 and from gear 6 to gear 2.
- The first table will be used to determine the angular velocity of the arm, and the second table for calculating the angular velocity of the gear 2.
- Since gear 8 is fixed, its velocity is zero. Let the absolute angular velocity of arm be x. We start from gear 2 and end at gear 6

Table 1: from gear 6 to gear 8

	Gear 8	Gear 4	Gear 5	Gear 6	Arm 7
Motion with arm	x	x	x	x	x
Motion relative to arm	(600-x)(N_6/N_5) N_4/N_8)	(600-x)(N_6/N_5)	(600-x)(N_6/N_5)	600-x	0
Total Motion	x+(600-x)(N_6/N_5) N_4/N_8)				x

$$x + (600 - x)\left(\frac{N_6}{N_5}\right)\left(\frac{N_4}{N_8}\right) = 0 \qquad x = -1955 \text{ rpm}$$

40

Example 5.15 Solution cont;

Table 2: from gear 6 to gear 2

	Gear 2	Gear 3	Gear 4	Gear 5	Gear 6	Arm 7
Motion with arm	-1955	-1955	-1955	-1955	-1955	-1955
Motion relative to arm	$2555(N_6/N_5)(-N_4/N_3)(-N_3/N_2)$	$2555(N_6/N_5)(-N_4/N_3)$	$2555(N_6/N_5)$	$2555(N_6/N_5)$	2555	0
Total Motion	$-1955+2555(N_6/N_5)(-N_4/N_3)(-N_3/N_2)$					-1955

$$\omega_2 = -1955 + 2555\left(\frac{N_6}{N_5}\right)\left(-\frac{N_4}{N_3}\right)\left(-\frac{N_3}{N_2}\right) = -1955 + 2555\left(\frac{N_6}{N_5}\right)\left(\frac{N_4}{N_2}\right) \qquad \omega_2 = -1955 + 2555\left(\frac{120}{30}\right)\left(\frac{22}{35}\right) \qquad \omega_2 = 4469 \text{ rpm}$$

41

Belt Drive



42

Belt Types

Shape	Figure	Joint	Centre Distance
Flat		Yes	No Upper limit
Round		Yes	No Upper limit
V		None	Limited
Timing		None	Limited

43

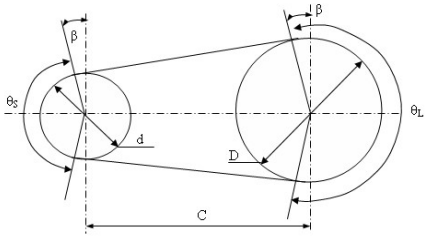
Consideration for Belt Drive Selection

1. Speed of the driving and driven shaft
2. Speed reduction ratio
3. Power to be transmitted
4. Center distance between shafts
5. Shaft layout

44

Length and Wrap Angle of Belt Drive

Open Belt

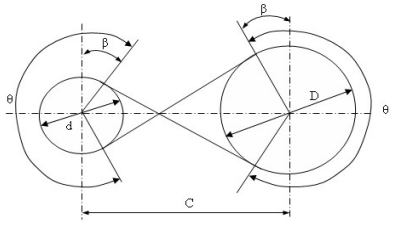


$$\beta = \sin^{-1}\left(\frac{D-d}{2C}\right) \qquad \theta_L = \pi + 2\beta$$

$$\theta_S = \pi - 2\beta$$

Length $$L = \sqrt{\left[4C^2 - (D-d)^2\right]} + \frac{1}{2}(D\theta_L + d\theta_S)$$

Crossed Belt

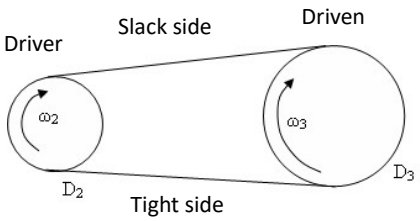


$$\beta = \sin^{-1}\left(\frac{D+d}{2C}\right) \qquad \theta = \pi + 2\beta$$

Length $$L = \sqrt{\left[4C^2 - (D-d)^2\right]} + \frac{1}{2}\theta(D+d)$$

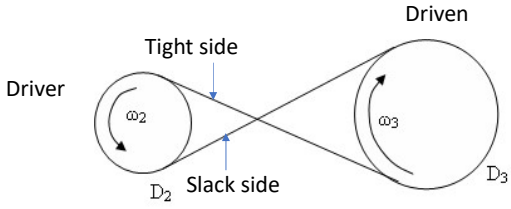
Belt Configurations: Non-reversing and Reversing belt drive

Non-reversing belt drive



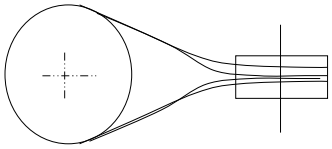
$$\frac{\omega_3}{\omega_2} = \frac{D_2}{D_3}$$

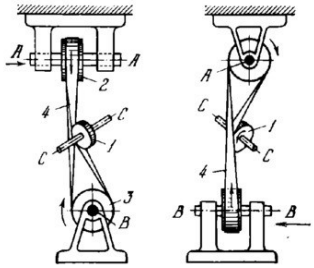
Reversing or Crossed belt drive

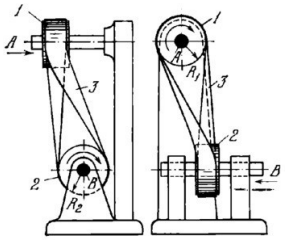


$$\frac{\omega_3}{\omega_2} = -\frac{D_2}{D_3}$$

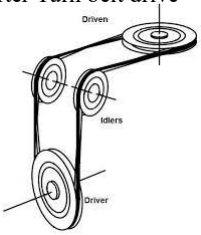
Belt Configurations: Quarter Turn belt drive





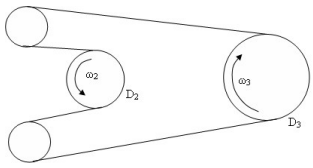


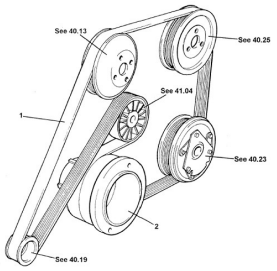
Quarter Turn belt drive



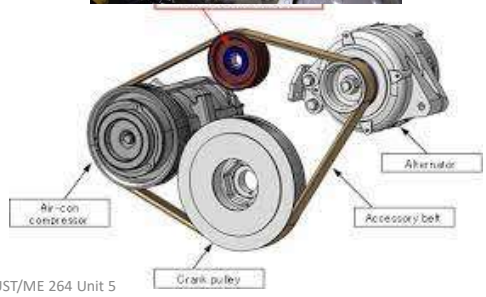
47

Belt Configurations:Multi-Driven Pulley Compound Driver



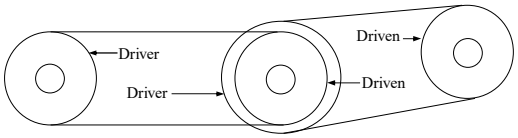


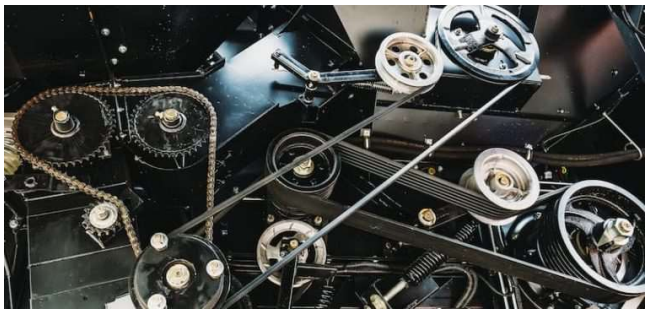


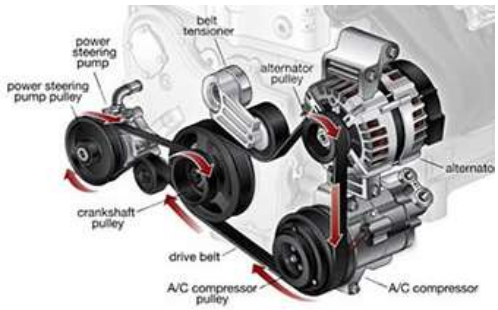


48

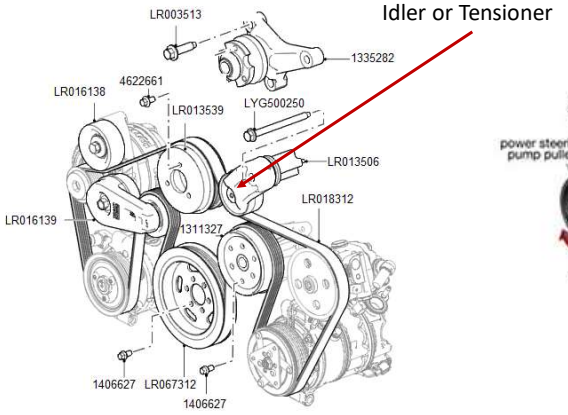
Belt Configurations: Multi-Stage Compound Pulley Driver

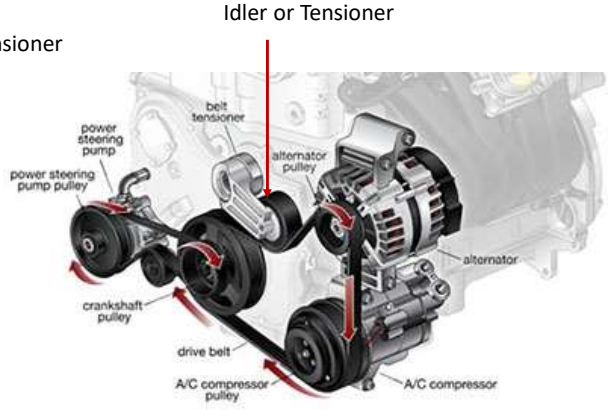






Belt Configurations: Idlers are used to adjust tension in belt and increase angle of wrap

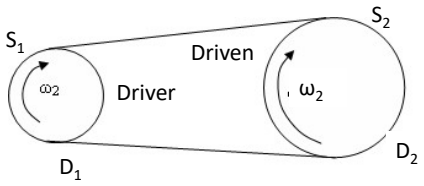




Belt Slip

- Sometimes, the frictional force which drive pulleys of a belt maybe insufficient. This may cause the belt to slip over the pulleys. This is called slip of belt and it is generally expresses as a percentage.
- Consider a belt drive with the following:
  - s_1 = percentage slip between the driver pulley and the belt;
  - s_2 = percentage slip between the belt and the driven pulley;
  - D_1 , D_2 = diameters of driver and driven pulleys, respectively
- Velocity of any point on the driver pulley is given by

$$v = \omega_1 D_1$$



Belt Slip

Velocity of a point on the belt is

$$V = \omega_1 D_1 - \omega_1 D_1 \frac{s_1}{100} \qquad V = \omega_1 D_1 \left(1 - \frac{s_1}{100}\right) \quad (1)$$

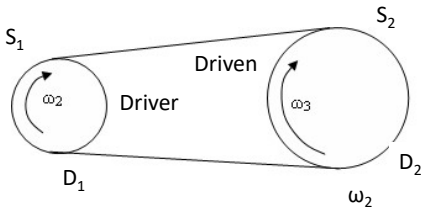
Similarly, the velocity of a point on the driven pulley is

$$\omega_2 D_2 = v - v\frac{s_2}{100} = \omega_1 D_1\left(\frac{s_1}{100}\right) - \omega_1 D_1\left(\frac{s_1}{100}\right)\frac{s_2}{100}$$

$$\omega_2 D_2 = v - v\frac{s_2}{100} = \omega_1 D_1\left(1 - \frac{s_1}{100}\right) - \omega_1 D_1\left(1 - \frac{s_1}{100}\right)\frac{s_2}{100} \quad (2)$$

Substituting (1) into (2), we have

$$\omega_2 D_2 = v - v\frac{s_2}{100} = \omega_1 D_1\left(1 - \frac{s_1}{100}\right)\left(1 - \frac{s_2}{100}\right)$$



$$\frac{\omega_2}{\omega_1} = \frac{D_1}{D_2}\left(1 - \frac{s_1}{100}\right)\left(1 - \frac{s_2}{100}\right) \quad (3)$$

If the thickness of the belt (t) is considered, then the Equation (3) becomes

$$\frac{\omega_2}{\omega_1} = \frac{D_1 + t}{D_2 + t}\left(1 - \frac{s_1}{100}\right)\left(1 - \frac{s_2}{100}\right) \quad (4)$$

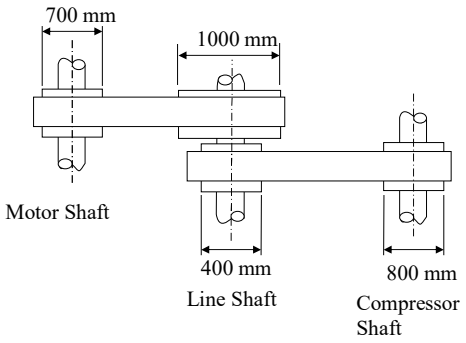
Example 5.16

A 3-phase electric motor is used to drive a reciprocating compressor through an intermediary line shaft, as shown. Knowing that the motor is running at 1500 rpm, determine the speed of the compressor shaft if

(a) there is no slip at all pulleys, and the thickness of each belts is 10 mm;

(b) there is 2% slip at each driver pulley, and the thickness of each belt is 10 mm;

(c) there are 2 % slip at each driver pulley, and 3 % slip at each driven pulley, thickness of first and second belts are 10 mm and 15 mm, respectively.



Example 5.16 Solution

Let the subscripts 1, 2, 3 and 4 denotes motor, driven line shaft, driver line shaft and compressor shaft, respective.

$$\frac{\omega_2}{\omega_1} = \frac{D_1 + t_f}{D_2 + t_f}\left(1 - \frac{s_1}{100}\right)\left(1 - \frac{s_2}{100}\right) \qquad \omega_2 = \omega_1 \frac{D_1 + t_f}{D_2 + t_f}\left(1 - \frac{s_1}{100}\right)\left(1 - \frac{s_2}{100}\right)$$

$$\frac{\omega_4}{\omega_3} = \frac{D_3 + t_s}{D_4 + t_s}\left(1 - \frac{s_3}{100}\right)\left(1 - \frac{s_3}{100}\right) \qquad \omega_4 = \omega_3 \frac{D_3 + t_s}{D_4 + t_s}\left(1 - \frac{s_3}{100}\right)\left(1 - \frac{s_4}{100}\right)$$

Since shafts 2 and 3 are same, $\omega_3 = \omega_2$

$$\omega_4 = \omega_1 \left(\frac{D_1 + t_f}{D_2 + t_f}\right)\left(\frac{D_3 + t_s}{D_4 + t_s}\right)\left(1 - \frac{s_1}{100}\right)\left(1 - \frac{s_2}{100}\right)\left(1 - \frac{s_3}{100}\right)\left(1 - \frac{s_4}{100}\right)$$

Example 5.16 Solution cont;

(a) $t_f = t_s = 10$ mm; $s_1 = s_2 = s_3 = s_4 = 0$;

$$\omega_4 = 1500\left(\frac{700+10}{1000+10}\right)\left(\frac{400+10}{800+10}\right)\left(1-\frac{0}{100}\right)\left(1-\frac{0}{100}\right)\left(1-\frac{0}{100}\right)\left(1-\frac{0}{100}\right) \qquad \omega_4 = 533.7 \text{ rpm}$$

(b) $t_f = t_s = 10$ mm; $s_1 = s_3 = 2\%$; $s_2 = s_4 = 0$;

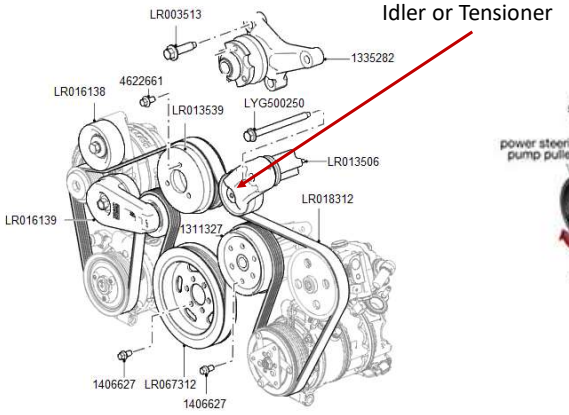
$$\omega_4 = 1500\left(\frac{700+10}{1000+10}\right)\left(\frac{400+10}{800+10}\right)\left(1-\frac{2}{100}\right)\left(1-\frac{0}{100}\right)\left(1-\frac{2}{100}\right)\left(1-\frac{0}{100}\right) \qquad \omega_4 = 512.6 \text{ rpm}$$

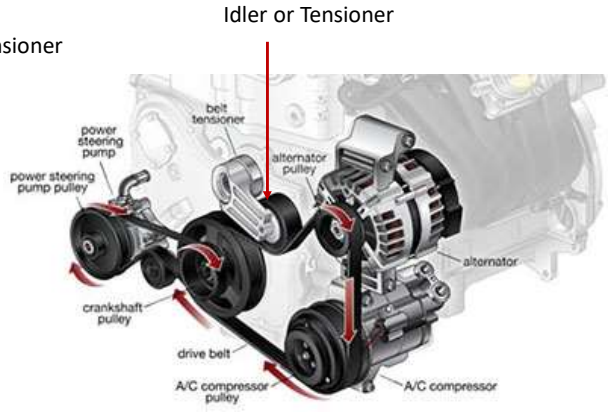
(c) $t_f = t_s = 10$ mm; $t_s = 15$ mm $s_1 = s_3 = 2\%$; $s_2 = s_4 = 3\%$;

$$\omega_4 = 1500\left(\frac{700+10}{1000+10}\right)\left(\frac{400+15}{800+15}\right)\left(1-\frac{2}{100}\right)\left(1-\frac{3}{100}\right)\left(1-\frac{2}{100}\right)\left(1-\frac{3}{100}\right) \qquad \omega_4 = 485.2 \text{ rpm}$$

55

Design of Multi-Driven Compound Belt Drive





56

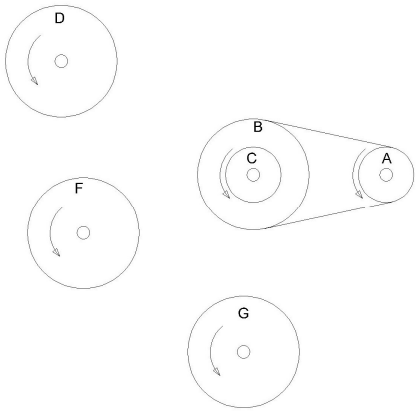
Example 5.17

Pulley A of an electric motor drives three pulleys D, F and G through an intermediary pulleys B and C, which are keyed to the same shaft. The diameters of the pulleys are: D_A = 100 mm, D_B = 400 mm, D_C = 100 mm, and D_D = 500 mm.

Assume that a single belt drives all the drive pulleys. Neglect thickness of the belt and effect of any driven pulley on another.

(a) Draw a single belt configuration over pulleys C, D, F and G such that

- they rotate in the directions shown;
- wrap angle on each driven pulley should be at least 160°;
- wrap angle on the driver pulley C should be at least 180°;
- You are allow to introduce maximum of two idlers to increase angle of wrap, each idler having diameter equal to half of diameter of driver pulley C.



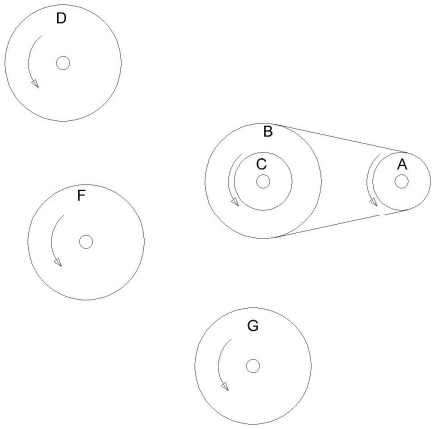
57

Example 5.17

(b) Knowing that pulley A is running at 1500 rpm, and there is 1.2% slip on driver pulleys A and C, and 1.4%, 2.5%, 2.0% and 1.6% slips on driven pulleys B, D, F and G respectively, find

i. the magnitudes of the angular speed of pulley D.

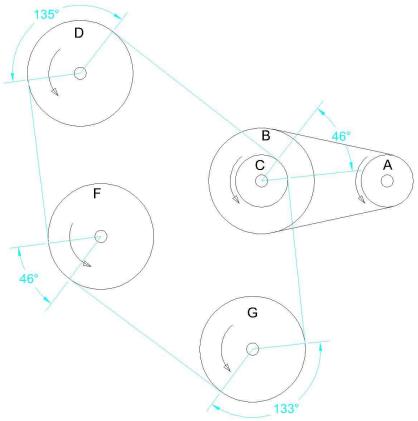
ii. the diameters of pulleys F and G such that they rotate at the same speed as that of pulley D.



58

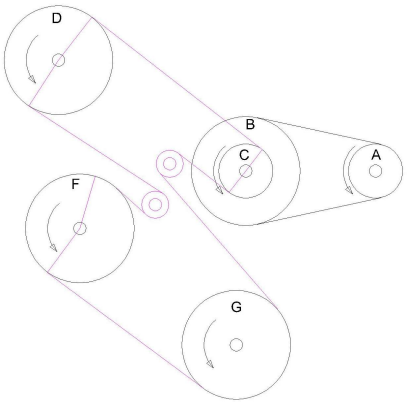
Example 5.17: Wrong Solution

- Normally belt is designed by drawing tangent lines from one pulley to another.
- Doing so may not yield the specified wrap angles, as shown.
- Use one idle pulley to increase angle of wrap on two adjacent pulleys.

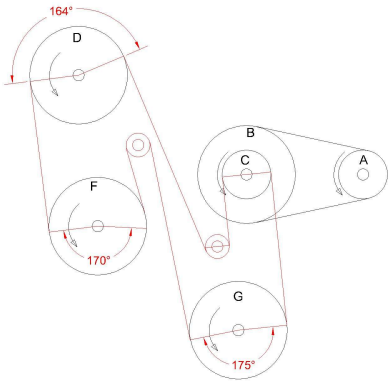


59

Example 5.17: Alternative solutions



Alternative Solution I



Alternative Solution II

60

Example 5.17 Calculation

ω_A =1500 rpm, s_A = s_C =1.5%, s_B =1.4%, s_D = 2.5%,

s_F =2.0% and s_G = 1.6%

D_A = 100; D_B = 400; D_C = 100; D_D=500

(b) $$\omega_H = \omega_A \left(\frac{D_A}{D_B}\right)\left(\frac{D_C}{D_H}\right)\left(1-\frac{s_A}{100}\right)\left(1-\frac{s_B}{100}\right)\left(1-\frac{s_C}{100}\right)\left(1-\frac{s_H}{100}\right)$$

(b)i where H = D in the above equation

$$\omega_D = 1500\left(\frac{100}{400}\right)\left(\frac{100}{500}\right)\left(1-\frac{1.5}{100}\right)\left(1-\frac{1.4}{100}\right)\left(1-\frac{1.5}{100}\right)\left(1-\frac{2.5}{100}\right) \qquad \omega_D = 70\ rpm$$

Example 5.17 Calculation

ω_A =1500 rpm, s_A = s_C =1.5%, s_B =1.4%, s_D = 2.5%, s_F =2.0% and s_G = 1.6%

D_A = 100; D_B = 400; D_C = 100; D_D=500

(b) $$\omega_H = \omega_A \left(\frac{D_A}{D_B}\right)\left(\frac{D_C}{D_H}\right)\left(1-\frac{s_A}{100}\right)\left(1-\frac{s_B}{100}\right)\left(1-\frac{s_C}{100}\right)\left(1-\frac{s_H}{100}\right)$$

(b)ii $$\omega_D = \omega_A \left(\frac{D_A}{D_B}\right)\left(\frac{D_C}{D_D}\right)\left(1-\frac{s_A}{100}\right)\left(1-\frac{s_B}{100}\right)\left(1-\frac{s_C}{100}\right)\left(1-\frac{s_D}{100}\right)$$

$$\omega_D=\omega_F = \omega_A \left(\frac{D_A}{D_B}\right)\left(\frac{D_C}{D_F}\right)\left(1-\frac{s_A}{100}\right)\left(1-\frac{s_B}{100}\right)\left(1-\frac{s_C}{100}\right)\left(1-\frac{s_F}{100}\right)$$

$$\Rightarrow\left(\frac{1}{D_D}\right)\left(1-\frac{s_D}{100}\right)=\left(\frac{1}{D_F}\right)\left(1-\frac{s_F}{100}\right) \qquad \left(\frac{1}{500}\right)\left(1-\frac{2.5}{100}\right)=\left(\frac{1}{D_F}\right)\left(1-\frac{2.0}{100}\right)$$

D_F=502.6 mm

Similarly $$\left(\frac{1}{500}\right)\left(1-\frac{2.5}{100}\right)=\left(\frac{1}{D_G}\right)\left(1-\frac{1.2}{100}\right)$$

D_G=504.6 mm